
Debian 13 (Trixie) - Security Overview

Dark theme English version

This document summarizes verifiable security changes in Debian 13 based on official release notes. The focus is on what was added, what was removed, and which security surfaces were reduced versus which risks remain.

Core concept

Debian 13 represents an architectural step forward: a modern kernel, clearer firmware separation, and the removal of legacy components. The result is a reduced attack surface, but with a higher dependency on correct firmware.

Key additions

- Linux kernel 6.12 (LTS)
- Improved KASLR and memory isolation
- Stronger protection against Spectre and Meltdown variants
- Better IOMMU and DMA protection
- Separate non-free-firmware
- Firmware is now isolated from the rest of the non-free software, which improves auditability and reduces implicit exposure to binary blobs.

Removed and deprecated

- Obsolete packages without active upstream maintenance
- Older cryptographic standards
- Parts of legacy i386 desktop support

Security impact

Debian 13 reduces several historical vulnerability classes: legacy TLS, outdated syscall paths, and insecure default settings. Newer toolchains such as OpenSSL 3.2, GCC 14, and systemd 256 remove entire classes of known CVE issues.

Remaining risks

- OEM BIOS and ACPI implementations
- Proprietary firmware that cannot be fully audited
- Proprietary GPU drivers, for example Nvidia

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